University of the Philippines Diliman
College of Science
National Institute of Physics

Physics 180 THV-1
Problem Set 6-7

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IV - BS Physics

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Problem 6.1

Find the **second-order** and the **third-order** terms in the transition matrix element T_{fi} of Fermi's golden rule.

Solution

Problem 7.1

The semi-major axis a and semi-major axis b of a prolate spheroid with eccentricity x are related through

$$b = \sqrt{1 - x^2}a \tag{2.1}$$

If the volume and surface area of the prolate spheroid are

$$V_{\text{prolate}} = \frac{4}{3}\pi ab^2 \tag{2.2}$$

$$S_{\text{prolate}} = 2\pi b\left(b + \frac{a\sin^{-1}(x)}{x}\right) \tag{2.3}$$

and defining $\varepsilon = \dfrac{1}{3}x^2$, show that the equations in slide 36 (reproduced below) hold for small values of x.

$$a_2 A^{\frac{2}{3}} \to a_2 A^{\frac{2}{3}}\left(1 + \frac{2}{5}\varepsilon^2\right) \tag{2.4}$$

$$a_3 \frac{Z^2}{A^{\frac{1}{3}}} = a_3 \frac{Z^2}{A^{\frac{1}{3}}}\left(1 - \frac{1}{5}\varepsilon 2\right) \tag{2.5}$$

Solution

References

Appendices

Code

No code was used for this problem set